

DATA-260 HW2

Rachandeep Kaur

Github Link- <https://github.com/rachandeepkaur/data260-7117.git>

Step 0: Personal Configuration

Value	Definition	Result
SID4	Last four digits of SJSU Student ID	7117
PORT_BASE	$8000 + (\text{SID4} \bmod 900)$	8817
PREFIX	"s" + SID4	s7117
SEED	SID4	7117
VERIFY_SEED	$260000 + \text{SID4}$	267117
DOMAIN_ID	$\text{SID4} \bmod 8$	5

Hardware: MacBook Pro, 16GB RAM

Model: qwen3:8b

Commit hash:

Part 1. HTML & CSS

- Style your HW1 entity list/form so it remains usable at 375px width.

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```

Local Restaurant Inspections

Facility Name / ID:

FA0206933 - YUMMY KITCHEN

Program Site Address:

1711 BRANHAM LN A9, SAN JOSE, CA

Submitter Email:

inspector@countyhealth.gov

Inspection Findings Summary:

Provide inspection observations and compliance details (must be > 25 characters)...

Program Element / Category:

-- Select Inspection Category --



☐ I agree to the terms and conditions.

Submit Inspection Report

- Provide visible loading, empty, and error states.

```

const getSubmissionCount = createSubmissionCounter();

// Toggles which of the three result states (empty / loading / error) is visible.
// Pass "success" once analysisResults has been populated to hide the other two.
const setResultState = (state) => {
  const emptyState = document.getElementById("emptyState");
  const errorState = document.getElementById("errorState");
  const loader = document.getElementById("loader");
  const loadingText = document.getElementById("loadingText");

  emptyState.hidden = state !== "empty";
  errorState.hidden = state !== "error";
  loader.hidden = state !== "loading";
  loadingText.hidden = state !== "loading";
};

// Renders the JSON returned by the FastAPI
const renderAnalysis = (publish) => {
  const container = document.getElementById("analysisResults");
  if (!container) return;

  container.innerHTML = "";

  const heading = document.createElement("h2");
  heading.textContent = "Agent Analysis";
  container.appendChild(heading);

  const fields = [
    ["Title", publish.title],
    ["Tags", publish.tags.join(", ")],
    ["Summary", publish.summary],
    ["Reviewer Approved", String(publish.approved_by_reviewer)]
  ];
};

```

```

const renderRecordsTable = () => {
  const recordsBody = document.getElementById("recordsBody");
  const needle = recordsState.query.trim().toLowerCase();
  const filtered = needle
    ? recordsState.records.filter(
        (r) =>
          r.facility_name.toLowerCase().includes(needle) ||
          r.site_address.toLowerCase().includes(needle)
      )
    : recordsState.records;

  recordsBody.innerHTML = "";

  if (filtered.length === 0) {
    const tr = document.createElement("tr");
    const td = document.createElement("td");
    td.colSpan = 3;
    td.textContent = recordsState.records.length === 0 ? "No records yet." : "No records match.";
    tr.appendChild(td);
    recordsBody.appendChild(tr);
    return;
  }

  filtered.forEach((record) => {
    const tr = document.createElement("tr");
    [record.id, record.facility_name, record.site_address].forEach((value) => {
      const td = document.createElement("td");
      td.textContent = value;
      tr.appendChild(td);
    });
    recordsBody.appendChild(tr);
  });
};

const loadRecords = async () => {
  try {
    const response = await fetch("/api/records");
    if (!response.ok) throw new Error(`GET /api/records failed: ${response.status}`);
    recordsState.records = await response.json();
    renderRecordsTable();
  } catch (error) {
    console.error("Failed to load records:", error);
    setRecordsStatus("Failed to load records.", true);
  }
};

```

Inspection Findings Summary:

Observed minor violations during a routine inspection at this facility today.

Program Element / Category:

ROUTINE HEALTH INSPECTION

☒ I agree to the terms and conditions.

Submit Inspection Report



Loading...

Inspection Records

Search

Clear

ID	Facility Name / ID	Site Address
1	FA0206933 - YUMMY KITCHEN	1711 BRANHAM LN A9, SAN JOSE, CA 95118
2	FA0198212 - GOLDEN WOK (RENAMED)	455 E SANTA CLARA ST, SAN JOSE, CA 95112
		200 S 1ST ST SAN JOSE

☒ I agree to the terms and conditions.

Submit Inspection Report

Form submitted, but the agent analysis request failed. Please try again.

Inspection Records

Search by facility name or site address

Search

Clear

ID	Facility Name / ID	Site Address
1	FA0206933 - YUMMY KITCHEN	1711 BRANHAM LN A9, SAN JOSE, CA 95118
2	FA0198212 - GOLDEN WOK	455 E SANTA CLARA ST, SAN JOSE, CA 95112

Failed to add record to the records table.

Update Record

Record ID:

New Name:

Update Record

No records yet. Submit an inspection above to see the agent analysis here.

Inspection Records

Search by facility name or site a

Search

Clear

ID	Facility Name / ID	Site Address
No records yet.		

Highest-ID record deleted.

Update Record

Record ID:

New Name:

Update Record

Delete Highest-ID Record

Delete Highest ID

Part 2. FastAPI - Backend on Port Base

1. Add a FastAPI backend on your PORT_BASE. Creates record and redirects to home view showing the updated list.

```
zsh: terminated uvicorn app:app --host 0.0.0.0 --port 8817
(.venv) (base) pc@pcs-MacBook-Pro web_application % uvicorn app:app --host 0.0.0.0 --port 8817
INFO: Started server process [14238]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8817 (Press CTRL+C to quit)
```

```
<!-- Requirement I.4.a: Required text input for primary field with autofocus & placeholder -->
<div>
  <label for="facility_name">Facility Name / ID:</label><br>
  <input
    type="text"
    id="facility_name"
    name="facility_name"
    placeholder="FA0206933 - YUMMY KITCHEN"
    autofocus
    required
  />
</div>
<br>

<!-- Requirement I.4.b: Required text input for secondary field with descriptive placeholder -->
<div>
  <label for="site_address">Program Site Address:</label><br>
  <input
    type="text"
    id="site_address"
    name="site_address"
    placeholder="1711 BRANHAM LN A9, SAN JOSE, CA 95118"
    required
  />
</div>
<br>

<!-- Requirement I.4.c: Required email input with descriptive placeholder -->
<div>
  <label for="inspector_email">Submitter Email:</label><br>
  <input
    type="email"
    id="inspector_email"
    name="inspector_email"
    placeholder="inspector@countyhealth.gov"
    required
  />
</div>
<br>
```

```

// Add this submission to the records store (Create) so it shows up
// in the table below, then refresh the table.
try {
  const addResponse = await fetch("/api/records", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ facility_name: facilityName, site_address: siteAddress }),
  });
  if (!addResponse.ok) throw new Error(`POST /api/records failed: ${addResponse.status}`);
  setRecordsStatus("Record added.");
  await loadRecords();
} catch (error) {
  console.error("Failed to add record to store:", error);
  setRecordsStatus("Failed to add record to the records table.", true);
}

```

```

1
2
3 @app.post("/api/records", response_model=RecordOut, status_code=201)
4 def create_record(payload: RecordIn) -> dict:
5     global _next_id
6     record = {"id": _next_id, **payload.model_dump()}
7     _records.append(record)
8     _next_id += 1
9     return record
10

```

Local Restaurant Inspections

Facility Name / ID:

FA1234567

Program Site Address:

1711 Branham Ln A9, San Jose, CA 95118

Submitter Email:

deeprachan973@gmail.com

Inspection Findings Summary:

Routine health inspection conducted. Handwashing stations were fully stocked with warm water, soap, and paper towels. Internal food storage temperatures were verified within safe limits (<41°F for cold storage, >135°F for hot holding). All food contact surfaces were properly cleaned and sanitized. No major or critical violations observed at the time of inspection.

Program Element / Category:

FOOD PREP / FOOD SVC OP 6-25 EMPLOYEES

☒ I agree to the terms and conditions.

Submit Inspection Report



Loading...

Inspection Records

Search by facility name or site address

Search

Clear

ID	Facility Name / ID	Site Address
1	FA0206933 - YUMMY KITCHEN	1711 BRANHAM LN A9, SAN JOSE, CA 95118
2	FA0198212 - GOLDEN WOK	455 E SANTA CLARA ST, SAN JOSE, CA 95112
3	FA1234567	1711 Branham Ln A9, San Jose, CA 95118

Record added.

-
2. **Update** — updates the record with ID 1 to new domain-appropriate values
→ redirect to home, show updated list.

```

<div class="records-actions">
  <h3>Update Record</h3>
  <form id="updateForm">
    <label for="update_id">Record ID:</label>
    <input type="number" id="update_id" name="id" min="1" step="1" required>
    <label for="update_facility_name">New Name:</label>
    <input type="text" id="update_facility_name" name="facility_name" required>
    <button type="submit">Update Record</button>
  </form>
</div>

```

```

document.getElementById("updateForm").addEventListener("submit", async (event) => {
  event.preventDefault();
  const form = event.target;
  const recordId = form.id.value;
  const payload = { facility_name: form.facility_name.value };

  try {
    const response = await fetch(`/api/records/${recordId}`, {
      method: "PUT",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify(payload),
    });
    if (!response.ok) throw new Error(`PUT /api/records/${recordId} failed: ${response.status}`);
    form.reset();
    setRecordsStatus(`Record ${recordId} updated.`);
    await loadRecords();
  } catch (error) {
    console.error(`Failed to update record ${recordId}:`, error);
    setRecordsStatus(`Failed to update record ${recordId} (does it exist?).`, true);
  }
});

```

```

@app.put("/api/records/{record_id}", response_model=RecordOut)
def update_record(record_id: int, payload: RecordUpdate) -> dict:
    for record in _records:
        if record["id"] == record_id:
            record["facility_name"] = payload.facility_name
            return record
    raise HTTPException(status_code=404, detail=f"Record with id {record_id} not found")

```

Inspection Records

Search

Clear

ID	Facility Name / ID	Site Address
1	The Urban Fork	1711 BRANHAM LN A9, SAN JOSE, CA 95118
2	FA0198212 - GOLDEN WOK	455 E SANTA CLARA ST, SAN JOSE, CA 95112

Record 1 updated.

Update Record

Record ID:

New Name:

Update Record

3. Delete — deletes the record with the highest ID → redirect to home, show updated list.

```
<div class="records-actions">  
  <h3>Delete Highest-ID Record</h3>  
  <button type="button" id="deleteHighestBtn">Delete Highest ID</button>  
</div>
```

```
document.getElementById("deleteHighestBtn").addEventListener("click", async () => {
  try {
    const response = await fetch("/api/records/highest", { method: "DELETE" });
    if (!response.ok) throw new Error(`DELETE /api/records/highest failed: ${response.status}`);
    setRecordsStatus("Highest-ID record deleted.");
    await loadRecords();
  } catch (error) {
    console.error("Failed to delete highest-ID record:", error);
    setRecordsStatus("Failed to delete highest-ID record.", true);
  }
});
```

```
@app.delete("/api/records/highest", response_model=Optional[RecordOut])
def delete_highest_record() -> Optional[dict]:
    if not _records:
        return None
    highest = max(_records, key=lambda r: r["id"])
    _records.remove(highest)
    return highest
```

Inspection Records

ID	Facility Name / ID	Site Address
1	FA0206933 - YUMMY KITCHEN	1711 BRANHAM LN A9, SAN JOSE, CA 95118
2	FA0198212 - GOLDEN WOK	455 E SANTA CLARA ST, SAN JOSE, CA 95112
3	FA1234567	1711 Branham Ln A9, San Jose, CA 95118

Highest-ID record deleted.

4. Search — search box matching against primary or secondary field → list filters to matches only.

```

<form id="searchForm" class="inline">
  <input type="text" id="searchInput" placeholder="Search by facility name or site address">
  <button type="submit">Search</button>
  <button type="button" id="clearSearchBtn">Clear</button>
</form>

```

```

const renderRecordsTable = () => {
  const recordsBody = document.getElementById("recordsBody");
  const needle = recordsState.query.trim().toLowerCase();
  const filtered = needle
    ? recordsState.records.filter(
        (r) =>
          r.facility_name.toLowerCase().includes(needle) ||
          r.site_address.toLowerCase().includes(needle)
      )
    : recordsState.records;

  recordsBody.innerHTML = "";

  if (filtered.length === 0) {
    const tr = document.createElement("tr");
    const td = document.createElement("td");
    td.colSpan = 3;
    td.textContent = recordsState.records.length === 0 ? "No records yet." : "No records match.";
    tr.appendChild(td);
    recordsBody.appendChild(tr);
    return;
  }
}

```

Inspection Records

ID	Facility Name / ID	Site Address
1	The Urban Fork	1711 BRANHAM LN A9, SAN JOSE, CA 95118

Part 3: Stateful Agent Graph

Step 2: Setting up the State

```
class AgentState(TypedDict):
    title: str
    content: str
    email: str
    strict: bool
    task: str
    llm: ModelClient
    planner_proposal: Optional[dict]
    reviewer_feedback: Optional[dict]
    turn_count: int
```

```
def planner_node(state: AgentState) -> dict:
    """Calls the LLM (via state['llm']) to draft tags + a summary."""
    user = (
        f'Task: {state["task"]}\n'
        f'Title: "{state["title"]}"\n'
        f'Submitter: {state["email"]}\n\n'
        f'Content:\n{state["content"]}'
    )
    draft: PlannerOutput = state["llm"].call(
        system=PLANNER_SYSTEM, user=user, output_format=PlannerOutput
    )
    return {"planner_proposal": draft.model_dump()}
```

```

def reviewer_node(state: AgentState) -> dict:
    """Calls the LLM to grade planner_proposal; also folds its own
    revised_output back into planner_proposal so a rejected draft is
    corrected before the next loop, instead of being re-reviewed unchanged."""
    system = REVIEWER_SYSTEM_STRICT if state["strict"] else REVIEWER_SYSTEM_LENIENT
    user = (
        f'Task: {state["task"]}\n'
        f'Title: "{state["title"]}"\n'
        f'Submitter: {state["email"]}\n\n'
        f'Content:\n{state["content"]}\n\n'
        f'Planner\'s draft (JSON):\n{json.dumps(state["planner_proposal"])}'
    )
    review: ReviewerOutput = state["llm"].call(
        system=system, user=user, output_format=ReviewerOutput
    )
    return {
        "reviewer_feedback": review.model_dump(),
        "planner_proposal": review.revised_output.model_dump(),
    }

```

```

6
7
8 def supervisor_node(state: AgentState) -> dict:
9     """Bookkeeping only - no LLM call, no routing decision."""
10    return {"turn_count": state["turn_count"] + 1}
11
12
13 def router_logic(state: AgentState) -> str:
14     """Reads state only; both conditional edges below reuse this function -
15     the graph wiring decides what "planner"/"reviewer" mean at each edge."""
16     feedback = state.get("reviewer_feedback")
17     if feedback is not None and feedback.get("approved"):
18         return END
19     if state["turn_count"] >= MAX_TURNS:
20         return END
21     if not state.get("planner_proposal"):
22         return "planner"
23     return "reviewer"
24

```

```
def build_graph():
    graph = StateGraph[AgentState, None, AgentState, AgentState](AgentState)
    graph.add_node("supervisor", supervisor_node)
    graph.add_node("planner", planner_node)
    graph.add_node("reviewer", reviewer_node)

    graph.set_entry_point("supervisor")

    # Supervisor -> (no proposal -> Planner | has proposal -> Reviewer)
    graph.add_conditional_edges(
        "supervisor",
        router_logic,
        {"planner": "planner", "reviewer": "reviewer", END: END},
    )
    graph.add_edge("planner", "supervisor")

    # Reviewer -> (issues -> back to Supervisor | no issues -> END)
    graph.add_conditional_edges(
        "reviewer",
        router_logic,
        {"planner": "supervisor", "reviewer": "supervisor", END: END},
    )

    return graph.compile()
```

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/langgraph_agent.py
Finished after 2 supervisor turn(s)
=====
{
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Health inspection"
    ],
    "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
  },
  "reviewer_feedback": {
    "approved": true,
    "feedback": "",
    "revised_output": {
      "tags": [
        "Cold-holding temperatures",
        "Handwashing signage",
        "Health inspection"
      ],
      "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
    }
  }
}
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/langgraph_agent.py
```

Running and Testing

```

def stream_agent_graph(
    client: ModelClient,
    submission: InspectionSubmission,
    *,
    strict: bool = True,
    task: str = "Summarize the inspection findings and extract 3 topical tags.",
) -> AgentState:
    """Same as run_agent_graph, but drives the graph with .stream() instead of
    .invoke() and prints each node's incremental update as it executes - this
    is what actually shows the Supervisor -> Planner/Reviewer -> Supervisor/END
    hops happening, one at a time, instead of only the final result."""
    initial_state = _initial_state(client, submission, strict=strict, task=task)
    app = build_graph()

    final_state: dict = dict[str, object](initial_state)
    for step_num, step in enumerate(dict[str, Any] | Any](app.stream(initial_state), start=1):
        for node_name, update in step.items():
            print(f"--- step {step_num}: {node_name} ---")
            print(json.dumps(update, indent=2, default=str))
            final_state.update(update)
        return final_state # type: ignore[return-value]

```

Agent output screenshots

```

--- step 1: supervisor ---
{
  "turn_count": 1
}
--- step 2: planner ---
{
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Health inspection"
    ],
    "draft_summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
  }
}
--- step 3: supervisor ---
{
  "turn_count": 2
}
--- step 4: reviewer ---
{
  "reviewer_feedback": {
    "approved": false,
    "feedback": "[TEST HOOK] Forcing rejection to verify the reject-loop: the summary (\\\"Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours.\\\") needs to explicitly mention the corrective action deadline.",
    "revised_output": {
      "tags": [
        "Cold-holding temperatures",
        "Handwashing signage",
        "Health inspection"
      ],
      "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
    }
  },
  "planner_proposal": null
}
--- step 5: supervisor ---
{
  "turn_count": 3
}
--- step 6: planner ---
{
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Corrective action deadline"
    ],
    "draft_summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
  }
}
--- step 7: supervisor ---
{
  "turn_count": 4
}
--- step 8: reviewer ---
{
  "reviewer_feedback": {
    "approved": false,
    "feedback": "[TEST HOOK] Forcing rejection to verify the reject-loop: the summary (\\\"Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours.\\\") needs to explicitly mention the corrective action deadline.",
    "revised_output": {
      "tags": [
        "Cold-holding temperatures",
        "Handwashing signage",
        "Corrective action deadline"
      ],
      "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
    }
  }
}

```

```

--- step 8: reviewer ---
{
  "reviewer_feedback": {
    "approved": false,
    "feedback": "[TEST HOOK] Forcing rejection to verify the reject-loop: the summary (\"Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours.\") needs to explicitly mention the corrective action deadline.",
    "revised_output": {
      "tags": [
        "Cold-holding temperatures",
        "Handwashing signage",
        "Corrective action deadline"
      ],
      "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
    }
  },
  "planner_proposal": null
}
--- step 9: supervisor ---
{
  "turn_count": 5
}
--- step 10: planner ---
{
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Corrective action deadline"
    ],
    "draft_summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
  }
}
--- step 11: supervisor ---
{
  "turn_count": 6
}
--- step 12: reviewer ---
{
  "reviewer_feedback": {
    "approved": true,
    "feedback": "",
    "revised_output": {
      "tags": [
        "Cold-holding temperatures",
        "Handwashing signage",
        "Corrective action deadline"
      ],
      "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
    }
  },
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Corrective action deadline"
    ],
    "summary": "Health inspection found cold-holding issues and missing handwashing signs. Manager fixed refrigerator, signage to be posted within 48 hours."
  }
}
=====
Finished after 6 supervisor turn(s)
=====
{
  "planner_proposal": {
    "tags": [
      "Cold-holding temperatures",
      "Handwashing signage",
      "Corrective action deadline"
    ]
  }
}

```

Part 4 - Output Schema and Loop Safety

1. Pydantic validation on Planner output

```
# Each tag must be a string 3-30 characters long (enforced per-item, not just list length).
Tag = Annotated[str, Field(min_length=3, max_length=30)]
```

```
class PlannerOutput(BaseModel):
    tags: List[Tag] = Field(
        ...,
        min_length=3,
        max_length=3,
        description="Exactly 3 tags, each a 3-30 character topical concept/theme",
    )
    draft_summary: str = Field(..., description="One sentence, at most 25 words")

    @field_validator("draft_summary")
    @classmethod
    def _draft_summary_word_limit(cls, value: str) -> str:
        word_count = len(value.split())
        if word_count > 25:
            raise ValueError(f"draft_summary must be at most 25 words, got {word_count}")
        return value
```

```
(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/tests/test_planner_validation.py
```

```
=====
Pydantic validation test: PlannerOutput
=====
```

```
[PASS] VALID: exactly 3 tags, 3-30 chars each, summary = 25 words
-> ACCEPTED: tags=['health inspection', 'food safety', 'facility compliance']
```

```
[PASS] INVALID: only 2 tags
-> REJECTED: List should have at least 3 items after validation, not 2
```

```
[PASS] INVALID: 4 tags
-> REJECTED: List should have at most 3 items after validation, not 4
```

```
[PASS] INVALID: tag too short (<3 chars)
-> REJECTED: String should have at least 3 characters
```

```
[PASS] INVALID: tag too long (>30 chars)
-> REJECTED: String should have at most 30 characters
```

```
[PASS] INVALID: summary is 26 words (over the 25-word limit)
-> REJECTED: Value error, draft_summary must be at most 25 words, got 26
```

```
=====
ALL CHECKS PASSED
=====
```

2. On validation failure

```
def planner_node(state: AgentState) -> dict:
    """Calls the LLM (via state['llm']) to draft tags + a summary.

    On a fresh run this is a first attempt from just title/content. On a
    retry after a rejection OR a schema validation failure, reviewer_feedback
    carries the prior critique – fold it in so the new draft actually
    addresses what was wrong, instead of the Planner blindly regenerating
    the same thing. (There's no separate "validation_error" field in
    AgentState – a caught ModelClientError is reported through the same
    reviewer_feedback slot a Reviewer rejection uses, since both cases mean
    the same thing to this node: "the last draft wasn't good enough, here's
    why, try again.")"""
    feedback = state.get("reviewer_feedback")
    feedback_note = ""
    if feedback and not feedback.get("approved") and feedback.get("feedback"):
        feedback_note = (
            f'\n\nA previous draft was rejected with this feedback – '
            f'address it in this new draft:\n{feedback["feedback"]}'
        )

    user = (
        f'Task: {state["task"]}\n'
        f'Title: "{state["title"]}"\n'
        f'Submitter: {state["email"]}\n\n'
        f'Content:\n{state["content"]}{feedback_note}'
    )
    try:
        draft: PlannerOutput = state["llm"].call(
            system=PLANNER_SYSTEM, user=user, output_format=PlannerOutput
        )
    except ModelClientError as exc:
        # Schema validation (exactly 3 tags, 3–30 chars each, <=25 words) kept failing even after ModelClientError
        # Don't let this crash the graph: report it like a rejection so the *next* Planner turn sees exactly
        # no router_logic's "no proposal -> planner" branch fires again – bounded by the same MAX_TURNS
        return {
            "reviewer_feedback": {
                "approved": False,
                "feedback": f"[SCHEMA VALIDATION FAILED] Your last output was rejected: {exc}",
            },
        }
    return {"planner_proposal": draft.model_dump()}

```

```

(.venv) (base) pc@pcs-MacBook-Pro data260-7117 % python code/tests/test_planner_validation_retry.py
=== Case: Planner fails validation twice, then succeeds – must NOT crash ===
supervisor -> turn_count=1
planner call #1: CAUGHT ModelClientError, fed back as reviewer_feedback
reviewer_feedback = {'approved': False, 'feedback': "[SCHEMA VALIDATION FAILED] Your last output was rejected: Model did not return valid 'PlannerOutput' JSON after 3 attempt(s).
st error: ValidationError('draft_summary must be at most 25 words, got 31')"}
supervisor -> turn_count=2
planner call #2: CAUGHT ModelClientError, fed back as reviewer_feedback
reviewer_feedback = {'approved': False, 'feedback': "[SCHEMA VALIDATION FAILED] Your last output was rejected: Model did not return valid 'PlannerOutput' JSON after 3 attempt(s).
st error: ValidationError('draft_summary must be at most 25 words, got 31')"}
supervisor -> turn_count=3
planner call #3: SUCCEEDED
supervisor -> turn_count=4
reviewer ran -> approved=True

planner_calls: 4 reviewer_calls: 2
final approved: True
final turn_count: 2
planner calls that saw the validation-error feedback: [False, True, True, False]

PASS: a Planner that fails schema validation does not crash the graph – it retries with the error fed back.

=== Case: Planner NEVER produces valid output – must stop at MAX_TURNS, not loop forever ===
planner_calls: 7
final turn_count: 8 (MAX_TURNS=8)
final reviewer_feedback: {'approved': False, 'feedback': "[SCHEMA VALIDATION FAILED] Your last output was rejected: Model did not return valid 'PlannerOutput' JSON after 3 attempt(s).
ast error: ValidationError('draft_summary must be at most 25 words, got 31')"}

PASS: an always-failing Planner terminates at the turn ceiling instead of looping forever.

```

3. Fixed-input run (30 times)

```
reports > hw02 > cases > {} schema_input.json > ...
1  {
2    "facility_name": "FA0206933 - YUMMY KITCHEN",
3    "site_address": "1711 BRANHAM LN A9, SAN JOSE, CA 95118",
4    "inspector_email": "inspector@countyhealth.gov",
5    "program_element": "ROUTINE HEALTH INSPECTION",
6    "inspection_summary": "Observed improper cold-holding temperatures in the walk-in refrigerator and missing employ
7  }
8
```

reports > hw02 >  METRICS.md Preview Markdown

HW02 Metrics — Planner Schema-Validation Retry Study

Fixed input (reports/hw02/cases/schema_input.json , facility: "FA0206933 - YUMMY KITCHEN"), run through the real Supervisor/Planner/Reviewer LangGraph agent (code/langgraph_agent.py , qwen3:8b via Ollama) 30 times. Each run classified by how many times the Planner's raw output failed PlannerOutput Pydantic validation (exactly 3 tags, 3-30 chars each, <=25-word summary) before it produced schema-valid JSON, bounded by MAX_TURNS=8 . Source data: {} raw/schema_validation_runs.json ,  raw/schema_validation_runs.csv . Full console transcript:  RUN_LOG.txt .

Outcome table

Bucket	Count	Mean latency (ms)
valid first attempt	30	26314.1
valid after 1 retry	0	-
valid after 2plus retries	0	-
abandoned at ceiling	0	-

- Total runs: 30
- Runs where the Planner's first attempt was already schema-valid: 30/30
- Runs abandoned at the turn ceiling (Planner never produced valid JSON): 0/30

```

reports > hw02 > RUN_LOG.txt
1 [2026-09-14T21:07:30.211Z] Starting schema-validation retry study on fixed input (facility: 'FA0206933 - YUMMY KITCHEN'), 30 runs
2 [2026-09-14T21:07:55.026Z] run 1/30 (24775 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
3 [2026-09-14T21:08:22.131Z] run 2/30 (27104 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
4 [2026-09-14T21:08:48.577Z] run 3/30 (26446 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
5 [2026-09-14T21:09:13.805Z] run 4/30 (25227 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
6 [2026-09-14T21:09:39.136Z] run 5/30 (25330 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
7 [2026-09-14T21:10:04.391Z] run 6/30 (25255 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
8 [2026-09-14T21:10:30.605Z] run 7/30 (26214 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
9 [2026-09-14T21:10:58.063Z] run 8/30 (27457 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
10 [2026-09-14T21:11:23.195Z] run 9/30 (25131 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
11 [2026-09-14T21:11:49.595Z] run 10/30 (26400 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
12 [2026-09-14T21:13:35.471Z] run 11/30 (27501 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
13 [2026-09-14T21:12:43.715Z] run 12/30 (26618 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
14 [2026-09-14T21:13:10.110Z] run 13/30 (26394 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
15 [2026-09-14T21:13:35.471Z] run 14/30 (25360 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
16 [2026-09-14T21:13:59.723Z] run 15/30 (24251 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
17 [2026-09-14T21:14:24.321Z] run 16/30 (24598 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
18 [2026-09-14T21:14:48.808Z] run 17/30 (24486 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
19 [2026-09-14T21:15:13.190Z] run 18/30 (24381 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
20 [2026-09-14T21:15:37.756Z] run 19/30 (24565 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
21 [2026-09-14T21:16:03.424Z] run 20/30 (25668 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
22 [2026-09-14T21:16:28.780Z] run 21/30 (25356 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
23 [2026-09-14T21:16:55.577Z] run 22/30 (26796 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
24 [2026-09-14T21:17:22.293Z] run 23/30 (26715 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
25 [2026-09-14T21:17:47.865Z] run 24/30 (25572 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
26 [2026-09-14T21:18:17.243Z] run 25/30 (29378 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
27 [2026-09-14T21:18:49.023Z] run 26/30 (31779 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
28 [2026-09-14T21:19:15.448Z] run 27/30 (26424 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
29 [2026-09-14T21:19:42.661Z] run 28/30 (27213 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
30 [2026-09-14T21:20:14.078Z] run 29/30 (31416 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
31 [2026-09-14T21:20:39.691Z] run 30/30 (25612 ms): valid_first_attempt (planner_attempts=1, failures_before_success=0, final_turn_count=2, approved=False)
32 [2026-09-14T21:20:39.697Z] Done. Wrote raw/schema_validation_runs.json,csv and METRICS.md under /Users/pc/Documents/sjsu/Data-260-HW-GitHub/260-HW-Reports/
33 [2026-09-14T21:21:08.400Z] Starting turn-ceiling comparison on fixed input (facility: 'FA0206933 - YUMMY KITCHEN'), ceilings=[2, 10], 20 runs
34 [2026-09-14T21:21:08.439Z] === MAX_TURNS=2: 20 runs ===
35 [2026-09-14T21:21:20.085Z] ceiling=2 run 1/20 (11646 ms): HIT CEILING (final_turn_count=2, approved=False)
36 [2026-09-14T21:21:31.093Z] ceiling=2 run 2/20 (11007 ms): HIT CEILING (final_turn_count=2, approved=False)
37 [2026-09-14T21:21:42.553Z] ceiling=2 run 3/20 (11459 ms): HIT CEILING (final_turn_count=2, approved=False)
38 [2026-09-14T21:21:54.319Z] ceiling=2 run 4/20 (11765 ms): HIT CEILING (final_turn_count=2, approved=False)
39 [2026-09-14T21:22:06.009Z] ceiling=2 run 5/20 (11690 ms): HIT CEILING (final_turn_count=2, approved=False)
40 [2026-09-14T21:22:16.048Z] ceiling=2 run 6/20 (11028 ms): HIT CEILING (final_turn_count=2, approved=False)

```

4. Turn-ceiling comparison

```

5
6 def run_all(submission: InspectionSubmission) -> List[Dict[str, Any]]:
7     client = ModelClient()
8     results: List[Dict[str, Any]] = []
9
10    original_max_turns = lg.MAX_TURNS
11    try:
12        for ceiling in CEILINGS:
13            lg.MAX_TURNS = ceiling
14            app = lg.build_graph()
15            print(f"[{now_iso()}] === MAX_TURNS={ceiling}: {RUNS_PER_CEILING} runs ===")
16
17            for run in range(1, RUNS_PER_CEILING + 1):
18                started_at = now_iso()
19                record = run_one(app, client, submission, ceiling, run)
20                record["started_at"] = started_at
21                record["finished_at"] = now_iso()
22                results.append(record)
23    finally:
24        lg.MAX_TURNS = original_max_turns # restore, since this module stays imported
25
26    return results
27
28
29 def write_raw(results: List[Dict[str, Any]]) -> None:

```

```
34 [2026-09-14T21:21:08.439Z] === MAX_TURNS=2: 20 runs ===
35 [2026-09-14T21:21:20.085Z] ceiling=2 run 1/20 (11646 ms): HIT CEILING (final_turn_count=2, approved=False)
36 [2026-09-14T21:21:31.093Z] ceiling=2 run 2/20 (11007 ms): HIT CEILING (final_turn_count=2, approved=False)
37 [2026-09-14T21:21:42.553Z] ceiling=2 run 3/20 (11459 ms): HIT CEILING (final_turn_count=2, approved=False)
38 [2026-09-14T21:21:54.319Z] ceiling=2 run 4/20 (11765 ms): HIT CEILING (final_turn_count=2, approved=False)
39 [2026-09-14T21:22:06.009Z] ceiling=2 run 5/20 (11690 ms): HIT CEILING (final_turn_count=2, approved=False)
40 [2026-09-14T21:22:16.948Z] ceiling=2 run 6/20 (10938 ms): HIT CEILING (final_turn_count=2, approved=False)
41 [2026-09-14T21:22:27.984Z] ceiling=2 run 7/20 (11036 ms): HIT CEILING (final_turn_count=2, approved=False)
42 [2026-09-14T21:22:38.338Z] ceiling=2 run 8/20 (10353 ms): HIT CEILING (final_turn_count=2, approved=False)
43 [2026-09-14T21:22:48.901Z] ceiling=2 run 9/20 (10562 ms): HIT CEILING (final_turn_count=2, approved=False)
44 [2026-09-14T21:23:00.173Z] ceiling=2 run 10/20 (11272 ms): HIT CEILING (final_turn_count=2, approved=False)
45 [2026-09-14T21:23:10.687Z] ceiling=2 run 11/20 (10513 ms): HIT CEILING (final_turn_count=2, approved=False)
46 [2026-09-14T21:23:21.129Z] ceiling=2 run 12/20 (10442 ms): HIT CEILING (final_turn_count=2, approved=False)
47 [2026-09-14T21:23:31.935Z] ceiling=2 run 13/20 (10805 ms): HIT CEILING (final_turn_count=2, approved=False)
48 [2026-09-14T21:23:43.494Z] ceiling=2 run 14/20 (11559 ms): HIT CEILING (final_turn_count=2, approved=False)
49 [2026-09-14T21:23:54.396Z] ceiling=2 run 15/20 (10901 ms): HIT CEILING (final_turn_count=2, approved=False)
50 [2026-09-14T21:24:05.184Z] ceiling=2 run 16/20 (10787 ms): HIT CEILING (final_turn_count=2, approved=False)
51 [2026-09-14T21:24:15.851Z] ceiling=2 run 17/20 (10666 ms): HIT CEILING (final_turn_count=2, approved=False)
52 [2026-09-14T21:24:26.982Z] ceiling=2 run 18/20 (11131 ms): HIT CEILING (final_turn_count=2, approved=False)
53 [2026-09-14T21:24:37.957Z] ceiling=2 run 19/20 (10974 ms): HIT CEILING (final_turn_count=2, approved=False)
54 [2026-09-14T21:24:48.753Z] ceiling=2 run 20/20 (10796 ms): HIT CEILING (final_turn_count=2, approved=False)
55 [2026-09-14T21:24:48.760Z] === MAX_TURNS=10: 20 runs ===
56 [2026-09-14T21:25:14.466Z] ceiling=10 run 1/20 (25706 ms): COMPLETED (final_turn_count=2, approved=True)
57 [2026-09-14T21:25:39.371Z] ceiling=10 run 2/20 (24904 ms): COMPLETED (final_turn_count=2, approved=True)
58 [2026-09-14T21:26:05.676Z] ceiling=10 run 3/20 (26305 ms): COMPLETED (final_turn_count=2, approved=True)
59 [2026-09-14T21:26:31.642Z] ceiling=10 run 4/20 (25965 ms): COMPLETED (final_turn_count=2, approved=True)
60 [2026-09-14T21:26:57.605Z] ceiling=10 run 5/20 (25962 ms): COMPLETED (final_turn_count=2, approved=True)
61 [2026-09-14T21:27:23.120Z] ceiling=10 run 6/20 (25514 ms): COMPLETED (final_turn_count=2, approved=True)
62 [2026-09-14T21:27:49.529Z] ceiling=10 run 7/20 (26409 ms): COMPLETED (final_turn_count=2, approved=True)
63 [2026-09-14T21:28:14.726Z] ceiling=10 run 8/20 (25196 ms): COMPLETED (final_turn_count=2, approved=True)
64 [2026-09-14T21:28:40.677Z] ceiling=10 run 9/20 (25951 ms): COMPLETED (final_turn_count=2, approved=True)
65 [2026-09-14T21:29:07.200Z] ceiling=10 run 10/20 (26522 ms): COMPLETED (final_turn_count=2, approved=True)
66 [2026-09-14T21:29:33.079Z] ceiling=10 run 11/20 (25878 ms): COMPLETED (final_turn_count=2, approved=True)
67 [2026-09-14T21:30:00.133Z] ceiling=10 run 12/20 (27053 ms): COMPLETED (final_turn_count=2, approved=True)
68 [2026-09-14T21:30:28.281Z] ceiling=10 run 13/20 (28148 ms): COMPLETED (final_turn_count=2, approved=True)
69 [2026-09-14T21:30:54.839Z] ceiling=10 run 14/20 (26558 ms): COMPLETED (final_turn_count=2, approved=True)
70 [2026-09-14T21:31:19.898Z] ceiling=10 run 15/20 (25058 ms): COMPLETED (final_turn_count=2, approved=True)
71 [2026-09-14T21:31:47.358Z] ceiling=10 run 16/20 (27460 ms): COMPLETED (final_turn_count=2, approved=True)
72 [2026-09-14T21:32:14.762Z] ceiling=10 run 17/20 (27403 ms): COMPLETED (final_turn_count=2, approved=True)
73 [2026-09-14T21:32:41.223Z] ceiling=10 run 18/20 (26460 ms): COMPLETED (final_turn_count=2, approved=True)
```

HW02 Turn-Ceiling Comparison

Fixed input (`reports/hw02/cases/schema_input.json`, facility: "FA0206933 - YUMMY KITCHEN"), same model settings, run through the real Supervisor/Planner/Reviewer LangGraph agent (`qwen3:8b` via Ollama) 20 times at each of `MAX_TURNS=2` and `MAX_TURNS=10` (40 runs total). "Completed" means the Reviewer genuinely approved before the ceiling was reached. Source data: `{}` `raw/turn_ceiling_runs.json`, `raw/turn_ceiling_runs.csv`.

Results

MAX_TURNS	Runs	Completed	Completion rate	Mean latency (all runs)	Mean latency (completed only)
2	20	0/20	0.0%	11015.1 ms	- ms
10	20	20/20	100.0%	26437.8 ms	26437.8 ms

Recommendation

Pick `MAX_TURNS=10` for deployment. `MAX_TURNS=2` is not a viable ceiling at all — 0/20 completion isn't a marginal or model-dependent result, it's structurally guaranteed by this graph's shape. Tracing `router_logic`: the first Supervisor visit (`turn_count=1`) routes to the Planner; the *second* Supervisor visit (`turn_count=2`) is where the ceiling check (`turn_count >= MAX_TURNS`) runs *before* the "has proposal → Reviewer" branch. With `MAX_TURNS=2`, that check is already true, so the graph ends via the ceiling at the exact moment it would otherwise have routed to the Reviewer — the Reviewer never gets to run even once. This matches the data exactly: every single ceiling=2 run terminated at `final_turn_count=2` with `approved=False`, at a uniform ~11s (one Planner call, no Reviewer call at all).

The actual minimum viable ceiling for a single non-retry round is **3**, not 2 — once the Reviewer runs and approves, `router_logic`'s approval check short-circuits before the ceiling check, so the graph can end successfully at `turn_count=2` regardless of how high `MAX_TURNS` is set, as long as it's `>= 3`. That means the choice isn't really "2 vs. 10" in practice — it's "any ceiling ≥ 3 " vs. "2, which is broken by construction."

Given that `MAX_TURNS=10` vs. the bare minimum of 3 comes down to headroom at ceiling 10, all 20 runs still converged

5. Adversarial input and run it five times

reports > hw02 > cases > `{}` `adversarial_input.json` > ...

```
1  {
2    "facility_name": "FA0455812 - MULTI-VIOLATION DINER",
3    "site_address": "88 W SANTA CLARA ST, SAN JOSE, CA 95113",
4    "inspector_email": "inspector@countyhealth.gov",
5    "program_element": "ROUTINE HEALTH INSPECTION",
6    "inspection_summary": "Observed improper cold-holding temperatures in the walk-in refrigerator (47F, required 41F)",
7    "task": "Provide an exhaustive, fully detailed narrative summary that individually documents every single violation",
8  }
9
```

code > experiments >  run_adversarial_case_experiment.py > ...

```
118     ,
119
120
121 def run_all(submission: InspectionSubmission, task: str) -> List[Dict[str, Any]]:
122     client = ModelClient()
123     app = build_graph()
124     results: List[Dict[str, Any]] = []
125
126     for run in range(1, TOTAL_RUNS + 1):
127         started_at = now_iso()
128         record = run_one(app, client, submission, task, run)
129         record["started_at"] = started_at
130         record["finished_at"] = now_iso()
131         results.append(record)
132
133     return results
134
135
136 def write_raw(results: List[Dict[str, Any]]) -> None:
137     RAW_DIR.mkdir(parents=True, exist_ok=True)
138     (RAW_DIR / "adversarial_runs.json").write_text(json.dumps(results, indent=2))
139     with (RAW_DIR / "adversarial_runs.csv").open("w", newline="") as f:
140         writer = csv.DictWriter(f, fieldnames=CSV_FIELDS, extrasaction="ignore")
141         writer.writeheader()
142         writer.writerows(results)
143
144
```

reports > hw02 >  ADVERSARIAL_RUN_LOG.txt

```
1 [2026-09-14T21:47:35.763Z] Starting adversarial case study on /Users/pc/Documents/sjsu/Data-260-HW-Git/data260-7117
2 [2026-09-14T21:47:35.802Z] --- run 1/5: starting ---
3     turn=1 planner attempt #1: schema-valid
4     reviewer attempt #1: approved=False
5     turn=3 planner attempt #2: schema-valid
6     reviewer attempt #2: approved=False
7
```